Lecture #14 | 22 Oct. 2026

HW #3 due next Tuesday

Midterm on Nov 5 (Thursday)

Last time: "tool" that we use to calculate steady state error in response to a polynomial input

"The Type" is dependent on whether tracking or regulating

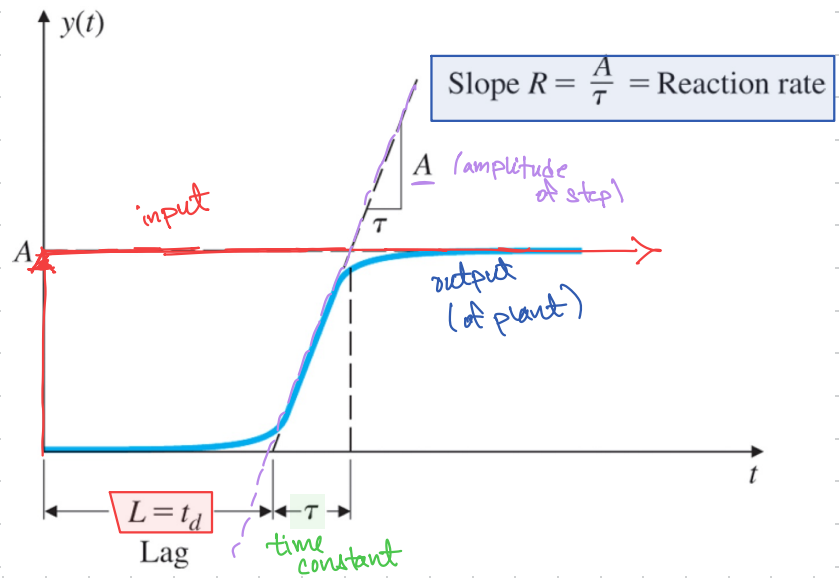
tracking: wrt. a reference

regulating: wrt a disturbance input

For tracking references, the system type is just the # of integrators in the product of the controller and plant

The major takeaway was: the system will tell you how to calculate the steady state error → just look it up in table.

- PID Tuning: "Simple Approaches" (compare to future: root locus, frequency response, Nyquist, etc)



Question: How do we tune a controller simply based off this information?

$$\frac{Y(s)}{U(s)} \approx \frac{A}{\tau s + 1} \cdot e^{-st_d}$$

simple 1st order, linear time invariant

time delay

← Model Representation

One Approach: Ziegler-Nichols method

$$D_c(s) = k_P\left(1 + \frac{1}{T_I s} + T_D s\right),$$

Note: definition is slightly different (but still equivalent)

with $k_I = k_P/T_I$ and $k_D = k_P T_D$. Table 4.2 (tuning for a decay ratio of 0.25):

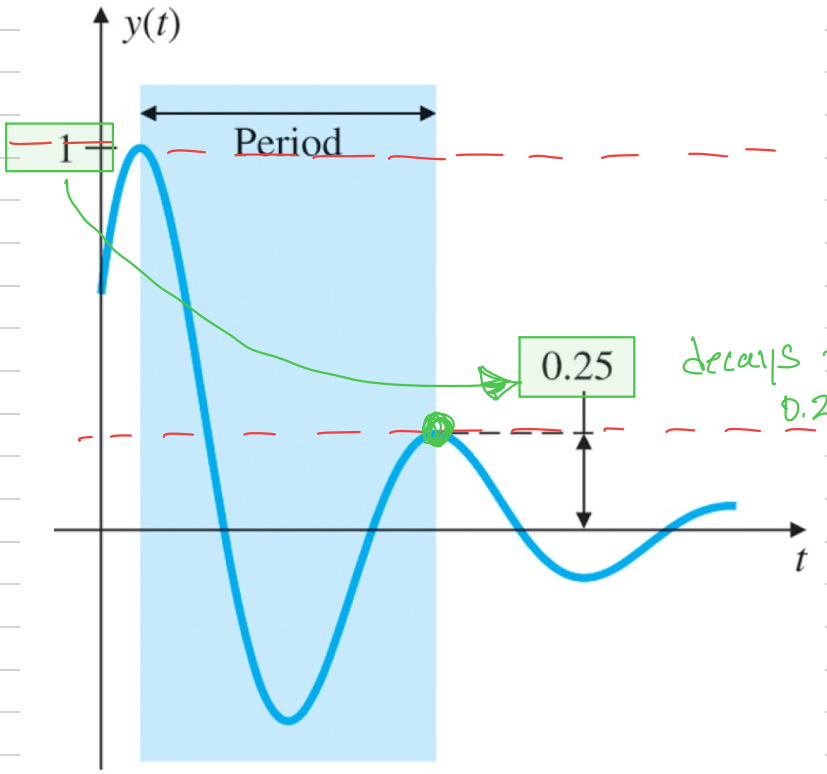
Controller	k_P	T_I	T_D
P	$\frac{1}{RL}$	–	–
PI	$\frac{0.9}{RL}$	$\frac{L}{0.3}$	–
PID	$\frac{1.2}{RL}$	$2L$	$0.5L$

If we wanted just prop. control

$\Rightarrow k_P = \frac{1}{RL}$

L is the "lag time"

R is the "reaction rate" (Δ/t)



decays to 0.25 in one period

The 2-N law was chosen so that the closed-loop transient decays to quarter of its amplitude in one period.

$M_p = \sqrt{1/4} = \frac{1}{2}$

$\Rightarrow \zeta = 0.2155$

the poles are pretty close to the imaginary axis, and consq. very close to the RHP.

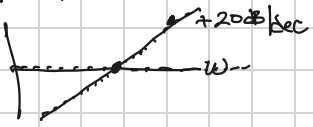
→ Establish closed-loop stability first, then start detuning.

P, I, D all have different effects on the system

P ↓ decrease, I ↓ decrease, D ↑ increase

↳ there is no free lunch → it amplifies measurement noise

derivative → a high pass filter



Sometimes derivative noise is significant enough that people will add lowpass filters to the derivative to attenuate the effect it has on measurement noise

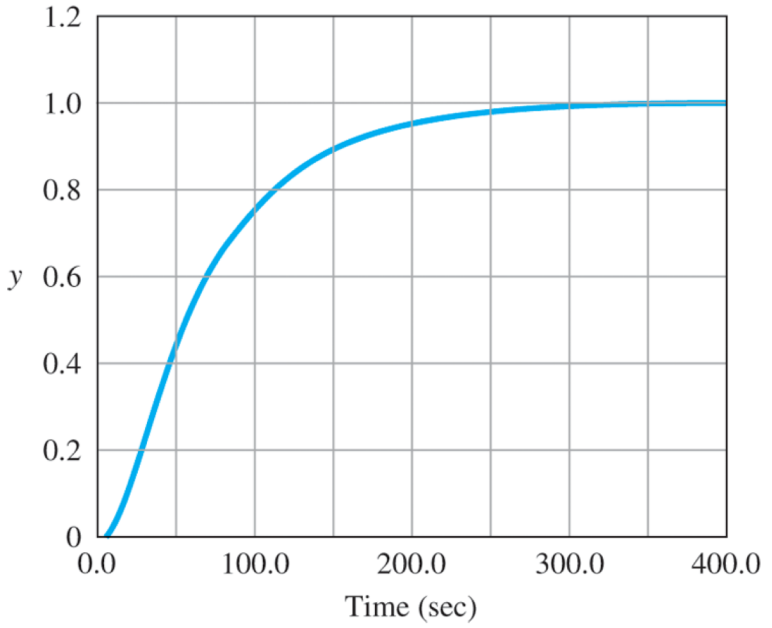
$$\left(\frac{K_D s^2 + K_P s + K_I}{s}\right) \cdot \frac{1}{s\tau + 1}$$

← PID w/ low pass filter

standard PID — Low pass filter

Example

PLANT



$R \approx 1/90\ 1/s$

$L \approx 13s$

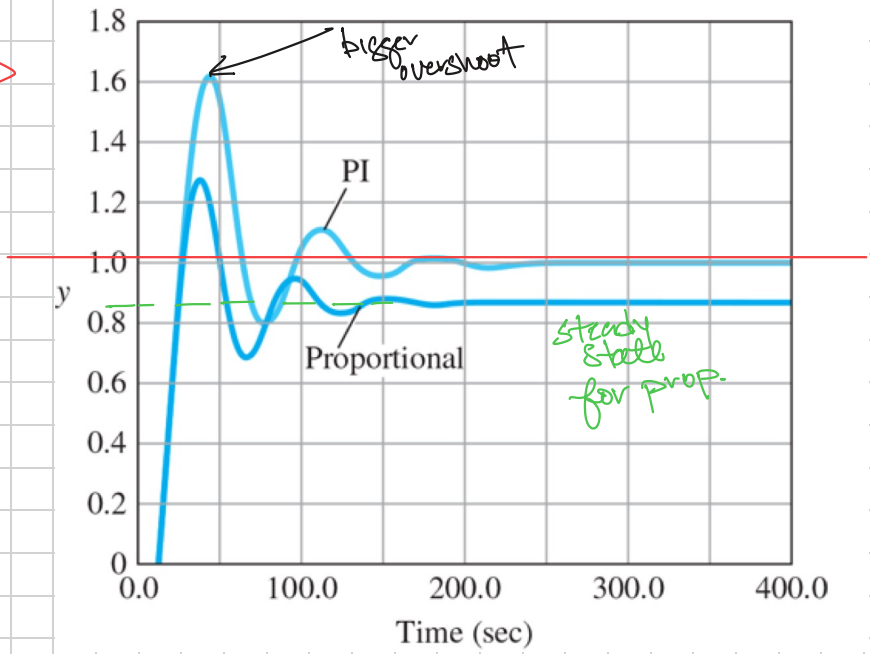
Z-N:

$P \Rightarrow K_P = \frac{1}{RL} = 6.92$

$PI \Rightarrow K_P = \frac{0.9}{RL} = 6.23$

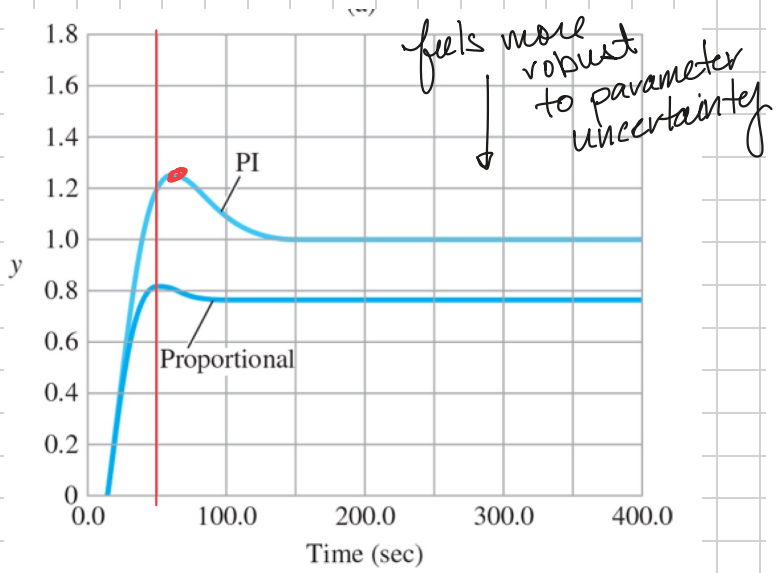
$T_i = \frac{L}{0.3} = 43.3s$

CLOSED LOOP



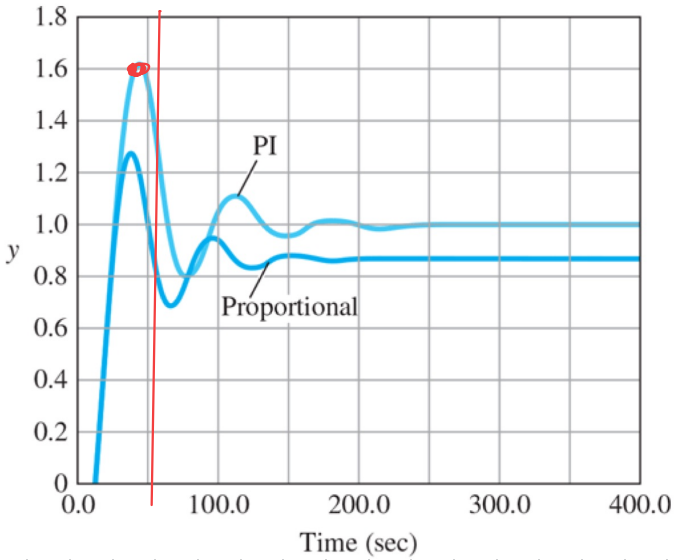
Overshoot - PI more
Rise Time - same
Settling Time - P sooner
SSE - Integral has zero SSE

Halve K_P



feels more robust to parameter uncertainty

Original



what did halving K_P do? →

Reduced settling time
Less overshoot
P has higher SSE

and rise time slowed down

The Ziegler method requires ability to measure the output.

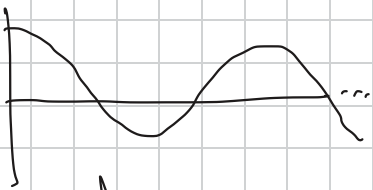
Sometimes you don't have that luxury

→ Ultimate Gain AND Ultimate Period

The method involves:

1) Create a Proportional Controller and close the loop
2) Crank up the P gain
3) displace the system from equilibrium
4) Observe the output
5) Does the system oscillate ($\zeta = 0$)
6) If it does, record the gain and the period of oscillation
   (K_u) (P_u)
7) If it doesn't oscillate

repeat



Z-N for "ultimate" gain and period

Controller	k_P	T_I	T_D
P	$0.5K_u$	–	–
PI	$0.45K_u$	$\frac{P_u}{1.2}$	–
PID	$0.6K_u$	$\frac{P_u}{2}$	$\frac{P_u}{8}$

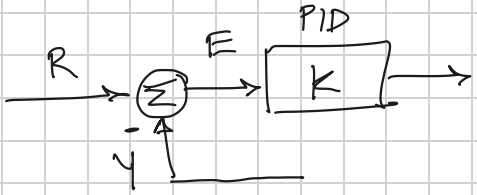
→ Same idea as before, but a different way of getting the coefficients
(low-fidelity, quick and dirty, sometimes works okay)

When does ZN not work:

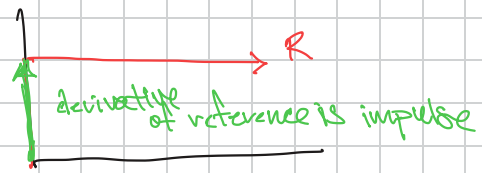
It works well for first order systems (stable)
It tends to not work well for higher order systems ($n \geq 3$)
It works terribly for unstable and/or nonminimum phase systems

e.g. aircraft: there is coupling b/n "angle-of-attack" and pitch
this is locally linear and first-order
SO, we can use Z-N laws to tune the pitch control of an aircraft.

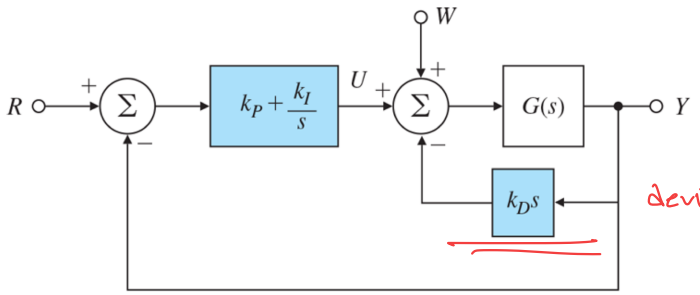
Derivative "kick"



Derivative acts on the reference signal, too



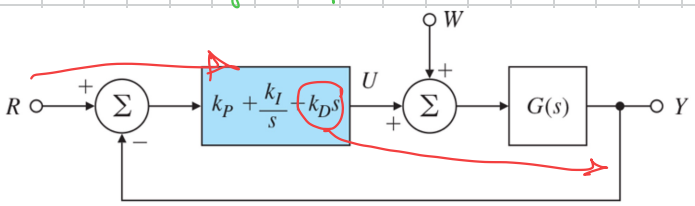
this physically unrealizable
→ the actuator will saturate
→ this causes "derivative kick"



(a)

derivative term ignores the reference signal
⇒ there's no "derivative kick"

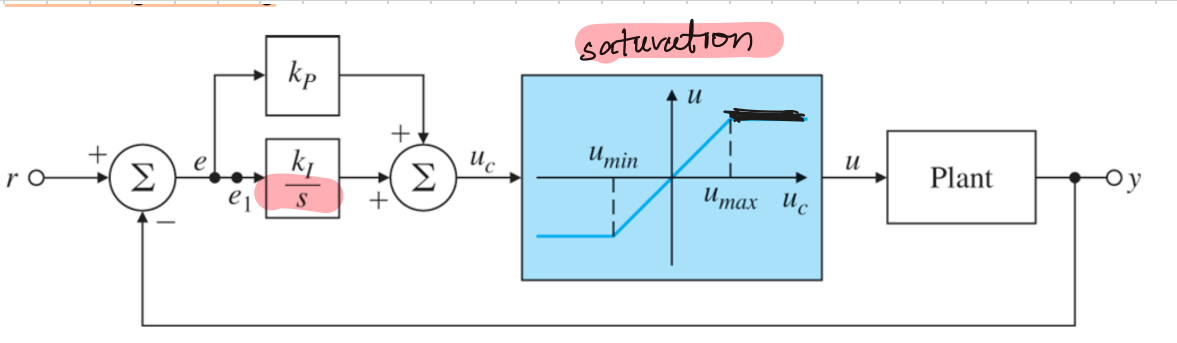
original pid



Trevor's recommendation: DO NOT feed step inputs into your reference

Command Profiling: what (reference) input do we apply to make sure the closed loop system behaves well.

Lasty Antiwindup: ① An integrator
② saturation on the plant's input



The integrator continues to accumulate error during saturated periods

When the system stops being saturated, the integrator has accumulated more error than it normally would have